

LAB# 5

Understanding the query by example (QBE) grid

The query design view is divided in two panes. The upper pane displays the tables you have selected for use in the query. The lower pane shows the query-by-example grid, in which you build the query or request.

Select queries

In Microsoft Access, the query is typically created by giving a symbolic example of the information to be retrieved. This technique is called Query-by-Example (QBE). It uses a special form called the QBE grid. In this section, an example is presented to show the general details of the approach.

The record type for the grade database is the following:

Student

SSN, First, Last, Street, City, State, Zip, Phone, Major, Class, Birthdate, Aid, Gender, HrsAttempted, HrsEarned, QualPoints

Grades

SSN, *Secid*, Grade

Sections

Secid, Cournum, Semester, Year, Instr

Course

Cournum, Courname, Credithr, Dept

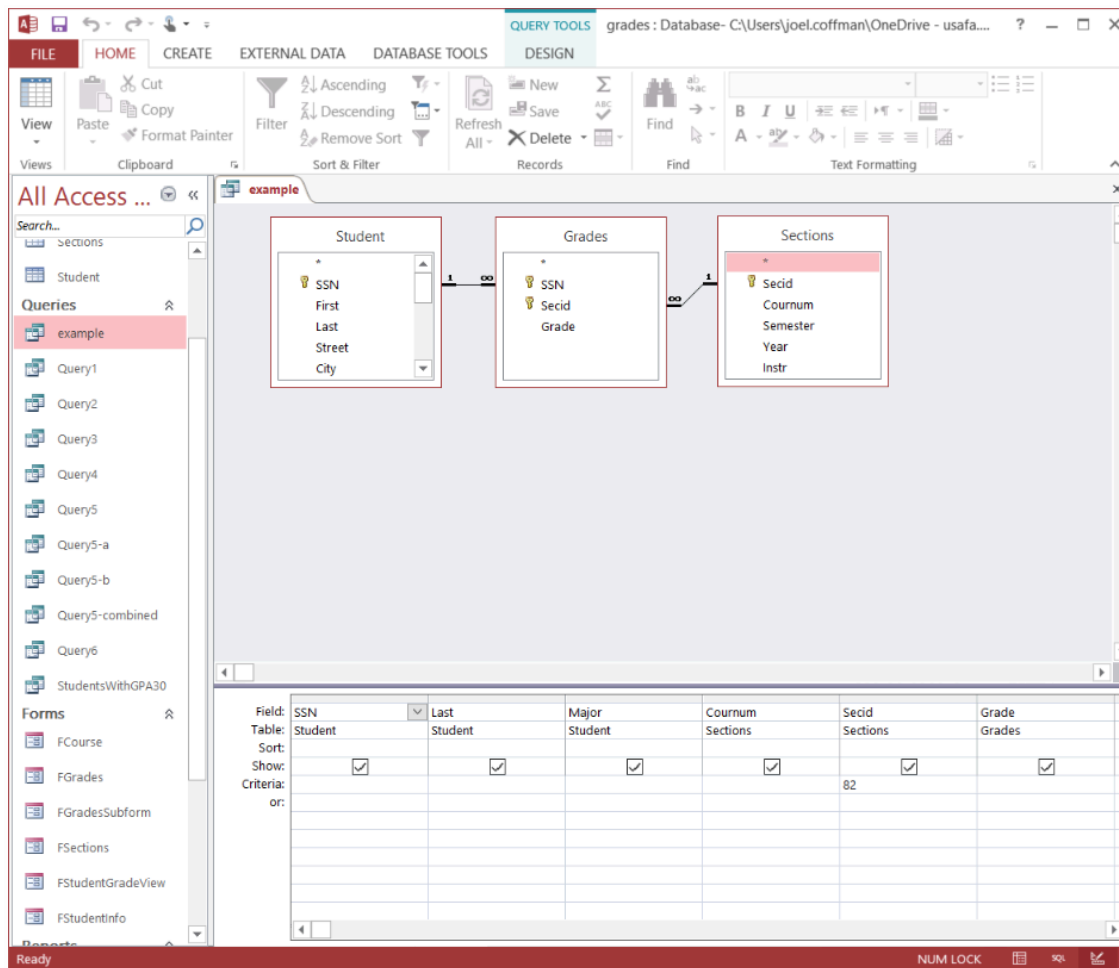
Where the primary key of each table is *italicized*. We must refer to this record type in creating a query.

Suppose that we need the following information from the database.

List the *SSN*, Last, Major, Cournum, *Secid*, and Grade for all students in section 82.

To carry out this request using Microsoft Access QBE, we begin by going to the Create ribbon.

Because we want to create a new query, we click the **Query Design** button.



On the initial display of the QBE grid, the grid may be partially hidden by the “Show Table” dialog box. A table is added to the upper portion of the QBE grid by either double clicking the name of the table or by highlighting the table name and clicking the **Add** button. We will need to add *Sections*, *Grades*, and *Student* to the grid.

In the QBE grid, field names to be used in the query are dragged from the tables in the top part to the field row in the grid. This can also be accomplished by double-clicking on the field names. For any field whose value is to be displayed, the “Show” box for that column must be clicked. In this case, all the fields are to be displayed. Add the fields SSN, Last, Major, Cournum, Secid, and Grade to the QBE grid

In the criteria row, enter the formula `=82` for the section identification field, Secid. This causes only those rows having 82 as the section id to be displayed. Criteria can be entered in this way for as many fields as needed. Fields having selection criteria can be included but not displayed

The following important rules hold for criteria on the QBE grid.

- Criteria in the *same* QBE grid row are combined using the **AND** operator.
- Criteria in *different* QBE grid rows are combined using the **OR** operator.

Clicking the Run button (exclamation point) in the Query Tools: Design ribbon causes the query to be executed with the result displayed in the Datasheet View window. You may toggle between the design view, the data view, and (this will become relevant soon) the SQL view by clicking the **View** button in the ribbon.

